

$$\|p\|_{\infty, X} \geq \beta \|p\|_{\infty} \quad \text{for all } p \in \mathcal{P}_2. \quad (5.24)$$

We choose for $p \in \mathcal{P}_2$ the monomial representation $p(x) = a_0 + a_1x + a_2x^2$. By evaluation of p on the point set $X = \{-1/2, 0, 1/2, 1\}$, we get

$$a_0 = p(0), \quad a_1 = p(1/2) - p(-1/2), \quad a_2 = p(1) - p(0) - p(1/2) + p(-1/2),$$

and therefore the (rough) estimate

$$\|p\|_{\infty} \leq |a_0| + |a_1| + |a_2| \leq \|p\|_{\infty, X} + 2\|p\|_{\infty, X} + 4\|p\|_{\infty, X} = 7\|p\|_{\infty, X}$$

for all $p \in \mathcal{P}_2$, whereby (5.24) holds for $\beta = 1/7$. Together with (5.23), this finally yields the sought estimate

$$\|p - f\|_{\infty} - \|p_2^* - f\|_{\infty} \geq \frac{1}{7} \|p - p_2^*\|_{\infty, X} \geq \frac{1}{49} \|p - p_2^*\|_{\infty} \quad \text{for all } p \in \mathcal{P}_2.$$

Therefore, p_2^* is the strongly unique best approximation to f . $\diamond$

5.3 Haar Spaces

In this section, we develop sufficient conditions for the approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$ under which a best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$ is strongly unique. To this end, we can rely on the result of Theorem 5.19, according to which we need to ensure the definiteness of $\|\cdot\|_{\infty, X}$ on $\mathcal{S}$ for any $X \subset E_{s^* - f}$.

We continue to use the assumptions and notations from the previous section, where $(s_1, \dots, s_n) \in \mathcal{S}^n$, for $n \in \mathbb{N}$, denotes an *ordered* basis of a finite-dimensional linear approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$. By the introduction of *Haar*³ spaces we specialize our assumptions on $\mathcal{S}$ and $(s_1, \dots, s_n)$ as follows.

Definition 5.22. *A linear space $\mathcal{S} \subset \mathcal{C}(\Omega)$ with $\dim(\mathcal{S}) = n < \infty$ is called a Haar space of dimension $n \in \mathbb{N}$ on Ω , if any $s \in \mathcal{S} \setminus \{0\}$ has at most $n - 1$ zeros on Ω . A basis $\mathcal{H} = (s_1, \dots, s_n) \in \mathcal{S}^n$ for a Haar space $\mathcal{S}$ on Ω is called a Haar system on Ω . $\bigcirc$*

In Haar spaces $\mathcal{S}$ of dimension $n \in \mathbb{N}$, we can solve interpolation problems for a discrete set $X \subset \Omega$ containing $|X| = n$ pairwise distinct points. In this case, $\|\cdot\|_{\infty, X}$ is a norm on $\mathcal{S}$, and so a solution of the interpolation problem is unique. We can further characterize Haar spaces as follows.

³ ALFRÉD HAAR (1885-1933), Hungarian mathematician

Theorem 5.23. *Let $\mathcal{S} \subset \mathcal{C}(\Omega)$ be a linear space of dimension $n \in \mathbb{N}$ and $X = \{x_1, \dots, x_n\} \subset \Omega$ a set of n pairwise distinct points. Then the following statements are equivalent.*

- (a) *Any $s \in \mathcal{S} \setminus \{0\}$ has at most $n - 1$ zeros on X .*
- (b) *For $s \in \mathcal{S}$, we have the implication*

$$s_X = 0 \quad \implies \quad s \equiv 0 \text{ on } \Omega,$$

i.e., $\|\cdot\|_{\infty, X}$ is a norm on $\mathcal{S}$.

- (c) *For any $f_X \in \mathbb{R}^n$, there is one unique $s \in \mathcal{S}$ satisfying $s_X = f_X$.*
- (d) *For any basis $\mathcal{H} = (s_1, \dots, s_n) \in \mathcal{S}^n$ of $\mathcal{S}$, the Vandermonde matrix*

$$V_{\mathcal{H}, X} = \begin{bmatrix} s_1(x_1) & \cdots & s_1(x_n) \\ \vdots & & \vdots \\ s_n(x_1) & \cdots & s_n(x_n) \end{bmatrix} \in \mathbb{R}^{n \times n}$$

is regular, where in particular $\det(V_{\mathcal{H}, X}) \neq 0$.

If one of the statements (a)-(d) holds for all sets $X = \{x_1, \dots, x_n\} \subset \Omega$ of n pairwise distinct points, then all of the remaining three statements (a)-(d) are satisfied for all X . In this case, $\mathcal{S}$ is a Haar space of dimension n on Ω .

Proof. Let $X = \{x_1, \dots, x_n\} \subset \Omega$ be a set of n pairwise distinct points in Ω . Obviously, the statements (a) and (b) are equivalent. By statement (b), the linear mapping $L_X : s \mapsto s_X$ is injective. Since $n = \dim(\mathcal{S}) = \dim(\mathbb{R}^n)$, this is equivalent to statement (c), i.e., L_X is surjective, and, moreover, also equivalent to statement (d), i.e., L_X is bijective. This completes our proof for the equivalence of statements (a)-(d).

If one of the statements in (a)-(d) holds for *all* sets $X = \{x_1, \dots, x_n\} \subset \Omega$ of n pairwise distinct points, then all of the remaining three statements in (a)-(d) are satisfied, due to the equivalence of statements (a)-(d). In this case, statement (a) holds in particular, for *all* sets $X = \{x_1, \dots, x_n\} \subset \Omega$, i.e., any $s \in \mathcal{S} \setminus \{0\}$ has at most $n - 1$ zeros on Ω , whereby $\mathcal{S}$ is, according Definition 5.22, a Haar space of dimension n on Ω . ■

According to the Mairhuber⁴-Curtis⁵ theorem [17, 48] there are *no* non-trivial Haar systems on *multivariate* connected domains $\Omega \subset \mathbb{R}^d$, $d > 1$. Before we prove the Mairhuber-Curtis theorem, we introduce a few notions.

Definition 5.24. *A domain $\Omega \subset \mathbb{R}^d$ is said to be **connected**, if for any pair of two points $x, y \in \Omega$ there is a continuous mapping $\gamma : [0, 1] \rightarrow \Omega$ satisfying $\gamma(0) = x$ and $\gamma(1) = y$, i.e., the points x and y can be connected by a continuous path in Ω . ○*

⁴ JOHN C. MAIRHUBER (1922-2007), US American mathematician

⁵ PHILIP C. CURTIS, JR. (1928-2016), US-American mathematician

Moreover, we call a domain $\Omega \subset \mathbb{R}^d$ **homeomorphic** to a subset of the sphere $\mathbb{S}^1 := \{x \in \mathbb{R}^2 \mid \|x\|_2 = 1\} \subset \mathbb{R}^2$, if for a non-empty and connected subset $U \subset \mathbb{S}^1$ there is a bijective continuous mapping $\varphi : \Omega \rightarrow U$ with continuous inverse $\varphi^{-1} : U \rightarrow \Omega$.

Theorem 5.25. (Mairhuber-Curtis, 1956/1959).

Let $\mathcal{H} = (s_1, \dots, s_n) \in (\mathcal{C}(\Omega))^n$ be a Haar system of dimension $n \geq 2$ on a connected set $\Omega \subset \mathbb{R}^d$, $d > 1$. Then, Ω contains no bifurcation, i.e., Ω is homeomorphic to a subset of the sphere $\mathbb{S}^1 \subset \mathbb{R}^2$.

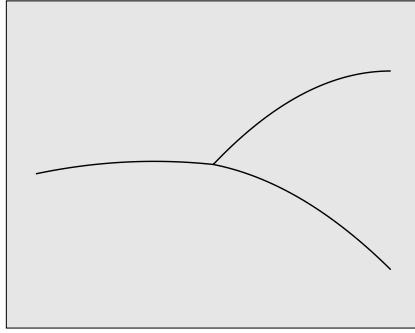


Fig. 5.3. According to the Mairhuber-Curtis theorem, Theorem 5.25, there are *no* non-trivial Haar systems $\mathcal{H}$ on domains Ω containing bifurcations.

Proof. Suppose Ω contains a bifurcation (see Figure 5.3 for illustration). Moreover, $X = \{x_1, \dots, x_n\} \subset \Omega$ be a subset of $n \geq 2$ pairwise distinct points in Ω . Now regard the determinant

$$d_{\{x_1, x_2, x_3, \dots, x_n\}} = \det(V_{\mathcal{H}, X}) = \det \begin{bmatrix} s_1(x_1) & s_1(x_2) & s_1(x_3) & \cdots & s_1(x_n) \\ s_2(x_1) & s_2(x_2) & s_2(x_3) & \cdots & s_2(x_n) \\ \vdots & \vdots & \vdots & & \vdots \\ s_n(x_1) & s_n(x_2) & s_n(x_3) & \cdots & s_n(x_n) \end{bmatrix}.$$

If $d_{\{x_1, x_2, x_3, \dots, x_n\}} = 0$, then $\mathcal{H}$, by Theorem 5.23, is not a Haar system.

Otherwise, we can shift the two points x_1 and x_2 by a *continuous* mapping along the two branches of the bifurcation, without any coincidence between points in X (see Figure 5.4).

Therefore, the determinant $d_{\{x_2, x_1, x_3, \dots, x_n\}}$ has, by swapping the first two columns in matrix $V_{\mathcal{H}, X}$, opposite sign to $d_{\{x_1, x_2, x_3, \dots, x_n\}}$, i.e.,

$$\operatorname{sgn}(d_{\{x_1, x_2, x_3, \dots, x_n\}}) = -\operatorname{sgn}(d_{\{x_2, x_1, x_3, \dots, x_n\}}).$$

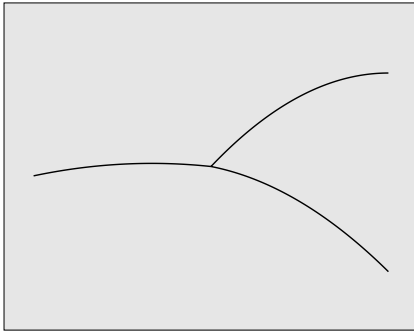
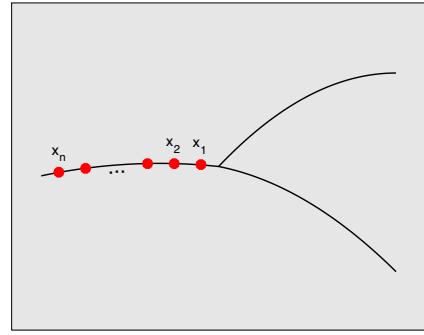
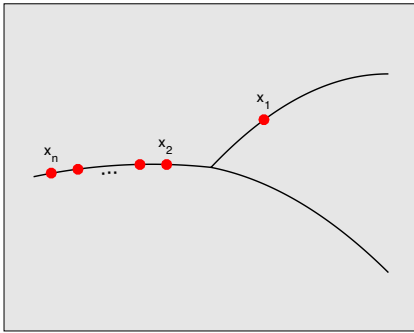
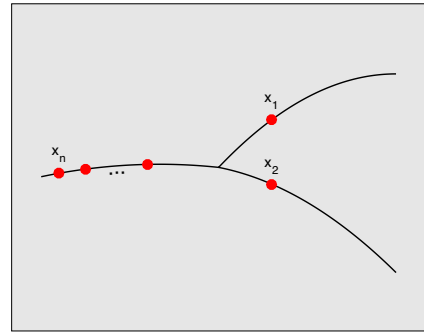
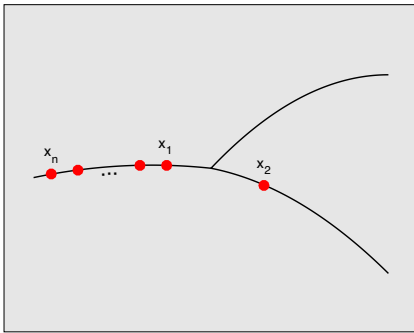
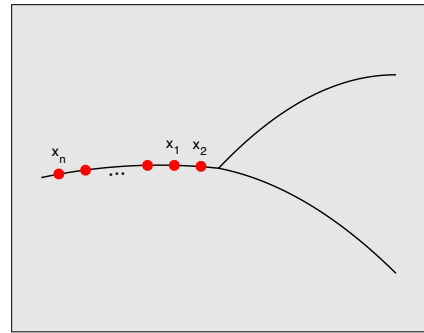

 domain Ω

 $X = (x_1, \dots, x_n) \in \Omega^n$

 step 1: shift of x_1

 step 2: shift of x_2

 step 3: back-shift of x_1

 step 4: back-shift of x_2

Fig. 5.4. Illustration of the Mairhuber-Curtis theorem, Theorem 5.25. The two points x_1 and x_2 can be swapped by a *continuous* mapping, i.e., by shifts along the branches of the bifurcation without coinciding with any other point from X .

Due to the continuity of the determinant, there must be a sign change of the determinant during the (continuous) swapping between x_1 and x_2 . In this case, $\mathcal{H} = \{s_1, \dots, s_n\}$ cannot be a Haar system, by Theorem 5.23. But this is in contradiction to our assumption to $\mathcal{H}$. ■

Due to the result of the Mairhuber-Curtis theorem, Theorem 5.25, we restrict ourselves from now to the *univariate* case, $d = 1$. Moreover, we assume from now that the domain Ω is a compact interval, i.e.,

$$\Omega = [a, b] \subset \mathbb{R} \quad \text{for } -\infty < a < b < \infty.$$

Before we continue our analysis on strongly unique best approximations, we first give a few elementary examples for Haar spaces.

Example 5.26. For $n \in \mathbb{N}_0$ and $[a, b] \subset \mathbb{R}$ the linear space of polynomials $\mathcal{P}_n$ is a Haar space of dimension $n+1$ on $[a, b]$, since according to the fundamental theorem of algebra any non-trivial polynomial from $\mathcal{P}_n$ has at most n zeros. ◇

Example 5.27. For $N \in \mathbb{N}_0$ the linear space $\mathcal{T}_N^{\mathbb{C}}$ of all complex trigonometric polynomials of degree at most N is a Haar space of dimension $N+1$ on $[0, 2\pi)$, since $\mathcal{T}_N^{\mathbb{C}}$ is, by Theorem 2.36, a linear space of dimension $N+1$, and, moreover, the linear mapping $p \mapsto p_X$, for $p \in \mathcal{T}_N^{\mathbb{C}}$ is, due to Theorem 2.39, for all sets $X \subset [0, 2\pi)$ of $|X| = N+1$ pairwise distinct points bijective.

Likewise, we can show, by using Corollaries 2.38 and 2.40, that the linear space $\mathcal{T}_n^{\mathbb{R}}$ of all real trigonometric polynomials of degree at most $n \in \mathbb{N}_0$ is a Haar space of dimension $2n+1$ on $[0, 2\pi)$. ◇

Example 5.28. For $[a, b] \subset \mathbb{R}$ and $\lambda_0 < \dots < \lambda_n$ the functions

$$\{e^{\lambda_0 x}, \dots, e^{\lambda_n x}\}$$

are a Haar system on $[a, b]$. We can show this by induction on n .

Initial step: For $n = 0$ the statement is trivial.

Induction hypothesis: Suppose the statement is true for $n-1 \in \mathbb{N}$.

Induction step ($n-1 \rightarrow n$): If a function of the form

$$u(x) \in \text{span} \{e^{\lambda_0 x}, \dots, e^{\lambda_n x}\}$$

has $n+1$ zeros in $[a, b]$, then the function

$$v(x) = \frac{d}{dx} (e^{-\lambda_0 x} \cdot u(x)) \quad \text{for } x \in [a, b]$$

has, according to the Rolle⁶ theorem, at least n zeros in $[a, b]$. However,

$$v(x) \in \text{span} \{e^{(\lambda_1 - \lambda_0)x}, \dots, e^{(\lambda_n - \lambda_0)x}\},$$

which implies $v \equiv 0$ by the induction hypothesis, and so $u \equiv 0$. ◇

⁶ MICHEL ROLLE (1652-1719), French mathematician

Example 5.29. The functions $f_1(x) = x$ and $f_2(x) = e^x$ are *not* a Haar system on $[0, 2]$. This is because $\dim(\mathcal{S}) = 2$ for $\mathcal{S} = \text{span}\{f_1, f_2\}$, but the continuous function

$$f(x) = e^x - 3x \neq 0$$

has by $f(0) = 1$, $f(1) = e - 3 < 0$ and $f(2) > 0$ at least two zeros in $[0, 2]$. Therefore, $\mathcal{S}$ cannot be a Haar space on $[0, 2]$. $\diamond$

Example 5.30. For $[a, b] \subset \mathbb{R}$ let $g \in \mathcal{C}^{n+1}[a, b]$ satisfy $g^{(n+1)}(x) > 0$ for all $x \in [a, b]$. Then, the functions $\{1, x, \dots, x^n, g\}$ are a Haar system on $[a, b]$: First note that the functions $1, x, \dots, x^n, g(x)$ are linearly independent, since from

$$\alpha_0 1 + \alpha_1 x + \dots + \alpha_n x^n + \alpha_{n+1} g(x) \equiv 0 \quad \text{for } x \in [a, b]$$

we can conclude $\alpha_{n+1} g^{(n+1)}(x) \equiv 0$ after $(n+1)$ -fold differentiation, whereby $\alpha_{n+1} = 0$. The remaining coefficients $\alpha_0, \dots, \alpha_n$ do also vanish, since the monomials $1, x, \dots, x^n$ are linearly independent. Moreover, we can show that any function $u \in \text{span}\{1, x, \dots, x^n, g\} \setminus \{0\}$ has at most $n+1$ zeros in $[a, b]$: Suppose

$$u(x) = \sum_{j=0}^n \alpha_j x^j + \alpha_{n+1} g(x) \neq 0$$

has $n+2$ zeros in $[a, b]$. Then, the $(n+1)$ -th derivative

$$u^{(n+1)}(x) = \alpha_{n+1} g^{(n+1)}(x)$$

has, due to the Rolle theorem, at least one zero in $[a, b]$. But this implies $\alpha_{n+1} = 0$, since $g^{(n+1)}$ is positive on $[a, b]$. In this case, $u \in \mathcal{P}_n$ is a polynomial of degree at most n , which, according to the fundamental theorem of algebra, vanishes identically on $[a, b]$. But this is in contraction to our assumption. $\diamond$

Now we return to the dual characterization of (strongly) unique best approximations. According to Corollary 5.15, there is for any best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{C}[a, b]$ a characterizing dual functional $\varphi : \mathcal{C}(\Omega) \rightarrow \mathcal{S}$ of the form

$$\varphi(u) = \sum_{j=1}^m \lambda_j \varepsilon_j u(x_j) \quad \text{for } u \in \mathcal{C}[a, b] \quad (5.25)$$

satisfying $\varphi(\mathcal{S}) = 0$, where $m \leq n+1$. For the case of Haar spaces $\mathcal{S} \subset \mathcal{C}[a, b]$ the length of the dual functional in (5.25) is necessarily $m = n+1$. Let us take note of this important observation.

Proposition 5.31. *Let $\varphi : \mathcal{C}[a, b] \rightarrow \mathbb{R}$ be a functional of the form (5.25), where $m \leq n+1$. Moreover, let $\mathcal{S} \subset \mathcal{C}[a, b]$ be a Haar space of dimension $\dim(\mathcal{S}) = n \in \mathbb{N}$ on $[a, b]$. If $\varphi(\mathcal{S}) = \{0\}$, then we have $m = n+1$.*

Proof. Suppose $m \leq n$. Then, due to Theorem 5.23 (c), the Haar space $\mathcal{S}$ contains one element $s \in \mathcal{S}$ satisfying $s(x_j) = \varepsilon_j$, for all $1 \leq j \leq m$. But for this s , we find $\varphi(s) = \|\lambda\|_1 = 1$, in contradiction to $\varphi(\mathcal{S}) = \{0\}$. $\blacksquare$

In the following discussion, we consider, for a fixed basis $\mathcal{H} = (s_1, \dots, s_n)$ of the Haar space $\mathcal{S}$, points $X = (x_1, \dots, x_{n+1}) \in I^{n+1}$, and sign vectors $\varepsilon = (\varepsilon_1, \dots, \varepsilon_{n+1}) \in \{\pm 1\}^{n+1}$, the *non-singular* Vandermonde matrices

$$V_{\mathcal{H}, X \setminus \{x_k\}} = \begin{bmatrix} s_1(x_1) & \cdots & s_1(x_{k-1}) & s_1(x_{k+1}) & \cdots & s_1(x_{n+1}) \\ \vdots & & \vdots & \vdots & & \vdots \\ s_n(x_1) & \cdots & s_n(x_{k-1}) & s_n(x_{k+1}) & \cdots & s_n(x_{n+1}) \end{bmatrix} \in \mathbb{R}^{n \times n} \quad (5.26)$$

for $1 \leq k \leq n+1$, and the *alternation matrix*

$$A_{\varepsilon, \mathcal{H}, X} = \begin{bmatrix} \varepsilon \\ V_{\mathcal{H}, X} \end{bmatrix} = \begin{bmatrix} \varepsilon_1 & \cdots & \varepsilon_{n+1} \\ s_1(x_1) & \cdots & s_1(x_{n+1}) \\ \vdots & & \vdots \\ s_n(x_1) & \cdots & s_n(x_{n+1}) \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}. \quad (5.27)$$

We first take note of a few properties for $V_{\mathcal{H}, X \setminus \{x_k\}}$ and $A_{\varepsilon, \mathcal{H}, X}$.

Proposition 5.32. *Let $\mathcal{H} = (s_1, \dots, s_n)$ be a Haar system on an interval $I \subset \mathbb{R}$ and $X = (x_1, \dots, x_{n+1}) \in I^{n+1}$ be a vector of $n+1$ pairwise distinct points. Then, the following statements are true.*

- (a) *For the Vandermonde matrices $V_{\mathcal{H}, X \setminus \{x_k\}}$ in (5.26), the signs of the $n+1$ determinants*

$$d_k = \det(V_{\mathcal{H}, X \setminus \{x_k\}}) \neq 0 \quad \text{for } 1 \leq k \leq n+1$$

are constant, i.e., $\text{sgn}(d_k) = \sigma$, for all $1 \leq k \leq n+1$, for some $\sigma \in \{\pm 1\}$.

- (b) *If the signs in $\varepsilon = (\varepsilon_1, \dots, \varepsilon_{n+1}) \in \{\pm 1\}^{n+1}$ are alternating, i.e., if*

$$\varepsilon_k = (-1)^{k-1} \sigma \quad \text{for } 1 \leq k \leq n+1$$

for some $\sigma \in \{\pm 1\}$, then the matrix $A_{\varepsilon, \mathcal{H}, X}$ in (5.27) is non-singular.

Proof. (a): Suppose we have $\text{sgn}(d_k) \neq \text{sgn}(d_{k+1})$ for some $1 \leq k \leq n$.

We consider a *continuous* mapping $\gamma : [0, 1] \rightarrow I$ satisfying $\gamma(0) = x_k$ and $\gamma(1) = x_{k+1}$. In this case, the *continuous* determinant mapping

$$d(\alpha) = \det(V_{\mathcal{H}, (x_1, \dots, x_{k-1}, \gamma(\alpha), x_{k+2}, \dots, x_{n+1})}) \quad \text{for } \alpha \in [0, 1]$$

satisfying $d(0) = d_{k+1}$ and $d(1) = d_k$ must have a sign change on $(0, 1)$. Due to the continuity of d there is one $\alpha^* \in (0, 1)$ satisfying $d(\alpha^*) = 0$. However, in this case, the Vandermonde matrix $V_{\mathcal{H}, (x_1, \dots, x_{k-1}, \gamma(\alpha^*), x_{k+2}, \dots, x_{n+1})} \in \mathbb{R}^{n \times n}$ is singular. Due to Theorem 5.23 (d), the elements in $(s_1, \dots, s_n)$ are *not* a Haar system on $I \subset \mathbb{R}$. But this is in contradiction to our assumption.

(b): According to the Laplace⁷ expansion (here with respect to the first row), the determinant of $A_{\varepsilon, \mathcal{H}, X}$ has the representation

$$\det(A_{\varepsilon, \mathcal{H}, X}) = \sum_{k=1}^{n+1} (-1)^{k+1} (-1)^{k-1} \sigma \cdot d_k = \sigma \sum_{k=1}^{n+1} d_k.$$

Due to statement (a), the signs of the determinants d_k , $1 \leq k \leq n+1$, are constant, which implies $\det(A_{\varepsilon, \mathcal{H}, X}) \neq 0$. $\blacksquare$

By using the results of Propositions 5.31 and 5.32, we can prove the *alternation theorem*, being the central result of this chapter. According to the alternation theorem, the signs $\varepsilon = (\varepsilon_1, \dots, \varepsilon_{n+1})$ of the dual characterization in (5.25) are for the case of Haar spaces $\mathcal{S}$ alternating. Before we prove the alternation theorem, we first give a formal definition for *alternation sets*.

Definition 5.33. Let $\mathcal{S} \subset \mathcal{C}(I)$ be a Haar space of dimension $n \in \mathbb{N}$ on an interval $I \subset \mathbb{R}$. Moreover, suppose $s^* \in \mathcal{S}$ and $f \in \mathcal{C}(I) \setminus \mathcal{S}$. Then, an ordered set $X = (x_1, \dots, x_{n+1}) \in E_{s^*-f}^{n+1} \subset I^{n+1}$ of $n+1$ monotonically increasing extremal points $x_1 < \dots < x_{n+1}$ is called an **alternation set** for s^* and f , if

$$\varepsilon_j = \operatorname{sgn}((s^* - f)(x_j)) = (-1)^j \sigma \quad \text{for all } j = 1, \dots, n+1$$

for some $\sigma \in \{\pm 1\}$, i.e., if the signs of $s^* - f$ are alternating on X . $\bigcirc$

Theorem 5.34. (Alternation theorem).

Let $\mathcal{S} \subset \mathcal{C}(I)$ be a Haar space of dimension $n \in \mathbb{N}$ on an interval $I \subset \mathbb{R}$. Moreover, let $I_K \subset I$ be a compact subset containing at least $n+1$ elements. Then, there is for any $f \in \mathcal{C}(I_K) \setminus \mathcal{S}$ a strongly unique best approximation $s^* \in \mathcal{S}$ to f with respect to $\|\cdot\|_{\infty, I_K}$. The best approximation s^* is characterized by the existence of an alternation set $X \in E_{s^*-f}^{n+1} \subset I_K^{n+1}$ for s^* and f .

Proof. Due to Corollary 3.8, any $f \in \mathcal{C}(I_K)$ has a best approximation $s^* \in \mathcal{S}$. Moreover, the strong uniqueness of s^* follows from Theorem 5.19, where the assumptions required therein for Theorem 5.17 are covered by Corollary 5.15.

Now we prove the stated characterization for s^* .

To this end, let $X = (x_1, \dots, x_{n+1}) \in E_{s^*-f}^{n+1} \subset I_K^{n+1}$ be an alternation set for s^* and f with (alternating) signs $\varepsilon_j = \operatorname{sgn}((s^* - f)(x_j)) = (-1)^j \sigma$, for $1 \leq j \leq n+1$, and some $\sigma \in \{\pm 1\}$. Then, we consider the linear system

$$\begin{bmatrix} \varepsilon_1 & \cdots & \varepsilon_{n+1} \\ s_1(x_1) & \cdots & s_1(x_{n+1}) \\ \vdots & & \vdots \\ s_n(x_1) & \cdots & s_n(x_{n+1}) \end{bmatrix} \cdot \begin{bmatrix} \varepsilon_1 \lambda_1 \\ \varepsilon_2 \lambda_2 \\ \vdots \\ \varepsilon_{n+1} \lambda_{n+1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (5.28)$$

⁷ PIERRE-SIMON LAPLACE (1749-1827), French mathematician and physicist

with the alternation matrix $A_{\varepsilon, \mathcal{H}, X}$ on the left hand side in (5.28). According to Proposition 5.32 (a), the matrix $A_{\varepsilon, \mathcal{H}, X}$ is non-singular. Therefore, the products $\varepsilon_k \lambda_k$, for $1 \leq k \leq n+1$, uniquely solve the linear system (5.28).

Due to the *Cramer*⁸ rule we have the representation

$$\varepsilon_k \lambda_k = \frac{(-1)^{k-1} d_k}{\det(A_{\varepsilon, \mathcal{H}, X})} \quad \text{for all } 1 \leq k \leq n+1,$$

where according to Proposition 5.32 (a) the signs of the $n+1$ determinants $d_k = \det(V_{\mathcal{H}, X \setminus \{x_k\}})$, for $1 \leq k \leq n+1$, are constant. This implies $\varepsilon_k \lambda_k \neq 0$, and, moreover, there is one unique vector $\lambda = (\lambda_1, \dots, \lambda_{n+1})^T \in A_{n+1}$ with *positive* coefficients

$$\lambda_k = \frac{d_k}{\sum_{j=1}^{n+1} d_j} > 0 \quad \text{for all } 1 \leq k \leq n+1$$

which solves the linear system (5.28). This solution $\lambda \in A_{n+1}$ of (5.28) finally yields the characterizing functional (according to Corollary 5.15),

$$\varphi(u) = \sum_{j=1}^{n+1} \lambda_j \varepsilon_j u(x_j) \quad \text{for } u \in \mathcal{C}(I_K), \quad (5.29)$$

satisfying $\varphi(\mathcal{S}) = \{0\}$. Due to Corollary 5.15, s^* is the (strongly unique) best approximation to f .

Now suppose that $s^* \in \mathcal{S}$ is the strongly unique best approximation to $f \in \mathcal{C}(I_K) \setminus \mathcal{S}$. Recall that the dual characterization in Corollary 5.15 proves the existence of a functional $\varphi : \mathcal{C}(I_K) \rightarrow \mathbb{R}$ of the form (5.25) satisfying $\varphi(\mathcal{S}) = \{0\}$, where φ has, according to Proposition 5.31, length $m = n+1$. We show that the point set $X = (x_1, \dots, x_{n+1}) \in E_{s^*-f}^{n+1}$ (from the dual characterization in Corollary 5.15) is an alternation set for s^* and f , where our proof is by contradiction. To this end, let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_{n+1}) \in \{\pm 1\}^{n+1}$ denote the sign vector of $s^* - f$ with $\varepsilon_j = \text{sgn}((s^* - f)(x_j))$, for $1 \leq j \leq n+1$. Now suppose there is one index $k \in \{1, \dots, n\}$ satisfying $\varepsilon_k = \varepsilon_{k+1}$. Then there is one $s \in \mathcal{S} \setminus \{0\}$ satisfying $s(x_j) = 0$ for all $j \notin \{k, k+1\}$ and $s(x_k) = \varepsilon_k$. Since s cannot have more than $n-1$ zeros in I , we necessarily have

$$\varepsilon_k = \text{sgn}(s(x_k)) = \text{sgn}(s(x_{k+1})) = \varepsilon_{k+1}.$$

This particularly implies

$$\varphi(s) = \lambda_k + \lambda_{k+1} |s(x_{k+1})| > 0,$$

which, however, is in contradiction to $\varphi(s) = 0$. ■

We finally remark that the characterizing alternation set $X \in E_{s^*-f}^{n+1}$ for s^* and f in the alternation theorem, Theorem 5.34, is *not* necessarily unique. This is because the set E_{s^*-f} of extremal points can be arbitrarily large (see Example 5.3).

⁸ GABRIEL CRAMER (1704-1752), Swiss mathematician